

What Do Vision-Language Models Really See?

Ordinary-Bench: Evaluating Spatial Belief via Controlled Probing and Scene Reconstruction

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Abstract

Vision-language models are increasingly evaluated on spatial reasoning benchmarks, yet question-level accuracy alone cannot tell us whether a model has formed a coherent and visually grounded belief about a scene. We introduce **Ordinary-Bench**, a controlled benchmark built from 700 procedurally generated 3D scenes and more than 336,000 ordinal spatial probes, spanning two task families: quaternary distance comparisons (QRR) and ternary directional judgments (TRR). Beyond local scoring, the benchmark provides *scene belief reconstruction*, which converts model responses into spatial constraints and evaluates the resulting scene-level belief in terms of satisfiability, geometric fidelity, and ambiguity, together with a three-condition protocol that contrasts correct-image, mismatched-image, and no-image settings to estimate visual information gain. Preliminary analyses indicate that models may achieve non-trivial local accuracy while still failing to externalize a globally coherent and visually grounded scene belief, with degradation becoming more pronounced as scenes grow more crowded. Ordinary-Bench therefore reframes VLM evaluation from “how many questions did the model answer correctly?” to “what kind of world does the model’s behavior imply it actually sees?”

1 Introduction

A vision-language model may correctly answer that the mug is left of the plate, that the spoon is closer than the fork, and that a ball moved behind a box. But if we interrogated the model the way a vision scientist interrogates an observer, would those answers add up to a single coherent percept of the world? This question matters because modern VLMs are increasingly asked to do more than recognize objects or describe images: they are evaluated on where objects are, how scenes are arranged, and what unfolds over time in videos. Yet success on question-answering benchmarks does not necessarily imply that a model has formed a stable spatial or temporal representation. A model may answer some questions correctly while relying on language priors, local appearance shortcuts, or mutually inconsistent heuristics. Recent work has made this tension increasingly visible: some benchmarks report progress on spatial reasoning, while others show that VLMs can “see” yet still fail to truly perceive, with responses heavily entangled with language priors [Chen et al. \[2024a\]](#), [Fu et al. \[2024a\]](#), [Chen et al. \[2024b\]](#), [Lee et al. \[2025\]](#). Similar concerns have emerged in video understanding [Fu et al. \[2024b\]](#), [Zhou et al. \[2024\]](#), [Liu et al. \[2024\]](#), [Zhou et al. \[2025\]](#). The problem, then, is not simply whether a model can answer spatial questions, but whether its answers reflect a coherent and visually grounded understanding of a world in space and time.

In biological vision research, progress rarely comes from a single aggregate score. It comes from controlled stimuli, dense behavioral probes, and carefully designed perturbations that reveal what an observer is sensitive to, what invariances it maintains, and what internal structure its behavior

implies. Current VLM evaluation is often much coarser. Most benchmarks score isolated responses: they tell us whether a model selected the right option, but not whether its answers can be assembled into a single geometrically plausible scene, whether that scene is grounded in the image, or how that belief changes when visual evidence is removed, mismatched, or altered. Accuracy alone therefore cannot distinguish seeing from guessing. Two models may achieve similar question-level scores while exhibiting radically different behavior: one may express a coherent and visually grounded scene belief, while the other may produce a collection of locally plausible but globally incompatible judgments. What is missing is not just another benchmark, but an evaluation framework that treats a VLM as a black-box observer and recovers the structure implied by its behavior.

We address this problem with **Ordinary-Bench**, a controlled benchmark for probing VLM spatiotemporal understanding in rendered 3D environments. Inspired by the diagnostic role of synthetic benchmarks such as CLEVR and CLEVRER [Johnson et al. \[2017\]](#), [Yi et al. \[2020\]](#), and more broadly by the logic of psychophysics and systems neuroscience [Kriegeskorte et al. \[2008\]](#), [Yamins and DiCarlo \[2016\]](#), Ordinary-Bench sacrifices ecological realism in exchange for exact geometric ground truth, controlled complexity, and repeatable perturbations. Our benchmark spans 700 procedurally generated scenes and more than 336,000 ordinal probes. Its static core consists of two task families: *Quaternary Relative Relations* (QRR), which compare distances between object pairs, and *Ternary Relative Relations* (TRR), which measure relative directional judgments. The framework also supports multi-view inputs and a temporal extension, but the central focus of this paper is a sharper question: when densely probed in a controlled world, what kind of scene belief does a VLM actually externalize?

To answer this question, Ordinary-Bench moves beyond question-level scoring and reconstructs the *externalized scene belief* that best explains a model’s responses. Model outputs are converted into spatial constraints, and the recovered scene is then evaluated for satisfiability, geometric fidelity, ambiguity, and distortion. This shift changes the unit of analysis. Rather than asking only how many local judgments are correct, we ask whether those judgments support a single coherent spatial explanation at all. Importantly, the goal is not to claim direct access to a model’s internal neural state. Instead, we adopt a form of behavioral reverse engineering: infer the structure of a model’s externalized scene belief from controlled probes, much as cognitive science uses structured tasks to study latent structure without equating behavior with mechanism. In this sense, Ordinary-Bench is both a benchmark and an evaluation methodology.

To separate genuine visual contribution from language priors, we complement this reconstruction framework with a three-condition protocol: correct image, mismatched image, and no image. Comparing behavior across these conditions allows us to estimate *visual information gain* and to test whether apparent spatial competence is driven by visual grounding or by textual priors alone. We further incorporate selective answering and consistency analysis, following recent work showing that abstention, uncertainty, and response stability are essential for reliable multimodal evaluation [Whitehead et al. \[2022\]](#), [Eisenschlos et al. \[2024\]](#), [Khan and Fu \[2024\]](#). Together, these mechanisms yield a stricter notion of competence: a strong model should not only answer some spatial questions correctly, but should do so in a way that is reconstructable, internally consistent, and causally tied to the image.

Our contributions are three-fold:

- We introduce **Ordinary-Bench**, a controlled benchmark for evaluating VLM spatial understanding in rendered 3D environments, with 700 scenes, over 336,000 ordinal probes, controlled complexity gradients, and support for multi-view and temporal extensions.
- We introduce **scene belief reconstruction** as a benchmark analysis tool: model responses

are converted into spatial constraints and used to recover an externalized scene belief, enabling scene-level metrics such as constraint satisfaction, geometry preservation, and reconstruction error beyond question accuracy.

- We introduce a **visual contribution separation** protocol based on correct-image, mismatched-image, and no-image conditions, together with selective answering and consistency analyses, to quantify how much of a model’s apparent spatial competence is genuinely grounded in vision.

Taken together, these components let us ask a sharper question than existing benchmarks typically permit: not merely whether VLMs can answer spatial questions, but what kind of world their behavior implies they actually see.

2 Related Work

Spatial benchmarks for VLMs. Recent work has shown that spatial reasoning remains a persistent weakness for vision-language models, even when their general multimodal scores appear strong. Dedicated benchmarks such as SpatialVLM [Chen et al. \[2024a\]](#), VSR [Liu et al. \[2023\]](#), and What’sUp [Kamath et al. \[2023\]](#) probe relative position, orientation, and spatial language understanding more directly than generic VQA datasets. However, these benchmarks still evaluate models primarily through question-level correctness. They tell us whether a response matches the label, but not whether a model’s responses can be assembled into a single scene-level geometric belief.

Controlled synthetic benchmarks. Ordinary-Bench is closer in spirit to controlled synthetic benchmarks such as CLEVR and CLEVRER [Johnson et al. \[2017\]](#), [Yi et al. \[2020\]](#), which trade ecological realism for exact ground truth and experimental control. This design choice is important for our setting: if the goal is to separate visual grounding from guessing and to test whether many local judgments imply a coherent spatial structure, then exact geometry, controlled complexity, and repeatable perturbations are more valuable than raw photorealism. We adopt this logic for dense ordinal probing rather than for compositional QA or event reasoning alone.

Language priors, hallucination, and evaluation mismatch. Several recent studies argue that apparent multimodal competence can be inflated by benchmark artifacts, textual shortcuts, or evaluation protocols that reward plausible language rather than genuine perception. BLINK highlights the gap between seeing and perceiving in multimodal models [Fu et al. \[2024a\]](#), MMStar questions whether current VLM evaluation pipelines actually measure the intended capabilities [Chen et al. \[2024b\]](#), POPE focuses on object hallucination [Li et al. \[2023\]](#), and VLInd-Bench explicitly measures language-prior contamination in VLM behavior [Lee et al. \[2025\]](#). Ordinary-Bench complements these efforts by moving from answer-level artifacts to scene-level structure: our question is not only whether a model guesses correctly without vision, but whether its answers remain coherent when interpreted as a spatial belief.

Selective answering and reliability. A second line of work emphasizes that reliable multimodal evaluation should account for uncertainty, abstention, and response consistency. Reliable VQA argues that abstaining is often preferable to confidently answering incorrectly [Whitehead et al. \[2022\]](#); Selectively Answering Visual Questions and the consistency-based selective VQA framework of Khan and Fu further show that uncertainty and response stability are essential to black-box

assessment Eisenschlos et al. [2024], Khan and Fu [2024]. Our protocol builds on this perspective. In addition to forced answering, Ordinary-Bench is designed to support selective-answering variants and to score structural consistency through transitivity and reciprocity, not just direct correctness.

Ordinal embedding and geometric reconstruction. The reconstruction component of Ordinary-Bench is grounded in work on ordinal embedding and geometric realizability. QRR constraints are naturally connected to ordinal embedding and non-metric multidimensional scaling Agarwal et al. [2007], Ma et al. [2018], while the question of whether a set of pairwise relations admits a valid geometric realization relates to Euclidean distance matrix theory Dokmanić et al. [2015]. TRR constraints are closer to bearing and angle rigidity, which characterize when directional or angular relations determine a configuration up to known symmetries Zhao and Zelazo [2019], Chen et al. [2021]. We do not present a new generic reconstruction algorithm; rather, we use these ideas to define a benchmark analysis tool that tests whether VLM responses imply any coherent scene at all.

3 Ordinary-Bench: Data & Protocol

Ordinary-Bench is a controlled benchmark for evaluating VLM spatial understanding through dense ordinal probing. It comprises 700 procedurally generated 3D scenes rendered in Blender, two families of ordinal spatial tasks (QRR and TRR) yielding over 336,000 probes in total, and a three-condition evaluation protocol that separates visual perception from language priors. This section describes each component.

3.1 Scene Generation

Scenes are generated using a Blender-based rendering pipeline. Each scene places n geometric primitives (spheres, cubes, cylinders) on a ground plane with randomized positions, sizes (`small`, `medium`, `large`), materials (`metal`, `rubber`), colors, and rotations. A minimum inter-object distance constraint ($d_{\min} = 0.25$ Blender units) prevents overlap. Lighting and camera parameters are fixed across scenes to control for visual confounds: the primary camera is positioned at azimuth 45° , elevation 30° , distance 12.0 units, rendering 480×320 images at 256 Cycles samples. Each scene additionally provides four viewpoints for multi-view experiments.

We generate seven *complexity splits*, indexed by object count $n \in \{4, 5, \dots, 10\}$. Each split contains 100 scenes produced from distinct random seeds, yielding $7 \times 100 = 700$ scenes in total. This design provides a controlled complexity gradient: the number of spatial relations grows combinatorially with n , letting us study how model performance degrades as scenes become more crowded. For each scene, the pipeline outputs (i) a rendered PNG image, (ii) a JSON file containing each object’s 3D coordinates, pixel coordinates, depth, and categorical attributes (shape, color, size, material), and (iii) multi-view images from four camera positions.

3.2 Task Definition

We define two families of ordinal spatial tasks that together probe both metric (distance) and directional (bearing) aspects of spatial understanding. Both are defined over ground-truth 3D coordinates, making them view-invariant.

Table 1: Probe counts per scene and per split (100 scenes each). QRR counts are before boundary filtering; actual counts are slightly lower due to tolerance-boundary exclusion.

n	QRR/scene	TRR/scene	Total/scene	Split total
4	3	24	27	2,700
5	15	60	75	7,500
6	45	120	165	16,500
7	105	210	315	31,500
8	210	336	546	54,600
9	378	504	882	88,200
10	630	720	1,350	135,000
Total				336,000

Quaternary Relative Relations (QRR). A QRR probe presents four objects organized as two pairs, (A, B) and (C, D) , and asks: *Is the 3D distance between A and B less than, approximately equal to, or greater than the distance between C and D?* The model must respond with one of three comparators: $<$, $\approx$, or $>$. Approximate equality is determined by a tolerance parameter $\tau = 0.10$: we define $d(A, B) \approx_{\tau} d(C, D)$ iff $|d(A, B) - d(C, D)| \leq \tau \cdot \max(d(A, B), d(C, D))$, with strict inequality otherwise. We restrict to *disjoint* pairs (no shared objects) to avoid trivially correlated judgments, and filter out boundary cases where the raw metric difference lies between 0.8τ and 1.2τ times the maximum distance (i.e., near the decision boundary) to ensure reliable labels. For n objects, the number of disjoint QRR probes is $\binom{n}{2}\binom{n-2}{2}/2$ before boundary filtering; for instance, a 10-object scene yields up to 630 QRR probes.

Ternary Relative Relations (TRR). A TRR probe specifies three objects—a *target*, a *reference origin* (**ref1**), and a *reference direction* (**ref2**)—and asks: *Standing at ref1 and facing toward ref2 (the 12 o’clock direction), at what clock hour (1–12) does the target appear?* The ground-truth answer is computed by projecting the target’s position onto the reference coordinate frame:

$$\alpha = (\text{atan2}(y_t - y_1, x_t - x_1) - \text{atan2}(y_2 - y_1, x_2 - x_1)) \bmod 360^{\circ}, \quad (1)$$

where subscripts $t, 1, 2$ denote target, **ref1**, **ref2** coordinates respectively. The angle α is then discretized into clock hours, each spanning a 30° sector. Evaluation uses two granularities: *hour-exact* (correct hour) and *quadrant-correct* (correct 90° quadrant). For n objects, TRR probes enumerate all ordered triples: $P(n, 3) = n(n-1)(n-2)$; a 10-object scene yields 720 TRR probes.

Probe counts. Table 1 summarizes the probe counts across complexity splits. The total benchmark contains 138,600 QRR probes and 197,400 TRR probes (before boundary filtering), yielding approximately 336,000 ordinal probes across all 700 scenes. The combinatorial growth from $n=4$ (27 probes/scene) to $n=10$ (1,350 probes/scene) provides a natural stress test of model capacity.

3.3 Evaluation Protocol

Batch JSON prompting. For each scene, probes are grouped into batches (default size 20) and presented to the VLM alongside the rendered image and a structured list of object descriptions. The model is instructed to return a JSON array of $\{\text{qid}, \text{answer}\}$ pairs. A ReAct-style correction loop re-prompts the model if the initial response contains parsing errors or missing answers. This batch format maximizes throughput while ensuring structured, machine-parseable outputs.

Three-condition protocol. To disentangle visual perception from language-driven guessing, we evaluate each model under three conditions on identical question sets:

- **Condition A** (correct image): the scene’s own rendered image is provided—the standard VQA setup.
- **Condition B** (mismatched image): a randomly selected image from a *different* scene is provided, so the visual input is valid but irrelevant to the questions.
- **Condition C** (no image): no image is provided; the model receives only textual object descriptions.

Comparing performance across conditions reveals the degree to which a model’s spatial answers are grounded in the visual input. We define the *Visual Information Gain* (VIG) as the improvement in reconstruction quality (measured by NRMS; see §4.3) from Condition C to Condition A: $VIG = NRMS_C - NRMS_A$. A positive VIG indicates that the correct image genuinely improves the model’s spatial belief; a near-zero VIG suggests the model relies primarily on language priors regardless of visual input.

Scoring. QRR probes are scored by exact comparator match. TRR probes are evaluated at two granularities: *hour-exact* (predicted hour matches ground truth) and *quadrant-correct* (predicted hour falls in the same 90° quadrant as the ground truth). We additionally track *adjacent* accuracy (predicted hour within ± 1 of ground truth, wrapping at 12 $\leftrightarrow$ 1) as a lenient measure. All metrics are reported per-split and aggregated.

3.4 Dataset Statistics and Release

Ordinary-Bench contains 700 scenes across 7 complexity splits, with 100 scenes each. Each scene includes a single-view rendered image, four multi-view images, and a JSON annotation file specifying object attributes and 3D/2D coordinates. The total dataset comprises approximately 336,000 ordinal probes (138,600 QRR + 197,400 TRR before boundary filtering). Scenes are partitioned into train and test subsets within each split to support model development without test-set contamination.

Object attributes span 3 shapes (sphere, cube, cylinder), 3 sizes (small, medium, large), 2 materials (metal, rubber), and 8 colors, placed on a ground plane ($z = 0$ for all objects). The procedural generation ensures full coverage of the complexity gradient while maintaining exact geometric ground truth. The release package is intended to include the dataset artifacts, generation code, evaluation scripts, question generation pipeline, and accompanying dataset documentation so that the full benchmark can be reproduced and extended.

4 Scene Belief Reconstruction

Question-level accuracy tells us how often a model’s answers match the ground truth, but not whether those answers are mutually consistent or whether they imply a coherent spatial layout. To move beyond per-question scoring, the benchmark provides a *scene belief reconstruction* pipeline that converts a model’s discrete responses into spatial constraints and solves for the 2D scene layout that best satisfies them. The resulting reconstruction—or its failure—reveals whether a model’s answers can be assembled into a geometrically plausible world.

4.1 From Responses to Constraints

Each model response to a QRR or TRR probe is converted into a spatial constraint over object positions.

Distance poset $\mathcal{P}_{\text{dist}}$. Each QRR response contributes an ordinal constraint on pairwise distances. For a probe comparing pairs (A, B) and (C, D) :

- Response “<” $\Rightarrow d(A, B) < d(C, D)$
- Response “ $\approx$ ” $\Rightarrow d(A, B) \approx d(C, D)$
- Response “>” $\Rightarrow d(A, B) > d(C, D)$

Pairs declared approximately equal are merged into equivalence classes via union-find, and strict orderings form directed edges. The resulting structure is a partially ordered set (poset) on distance equivalence classes, organized as a directed acyclic graph (DAG). A cycle in this DAG indicates a logical inconsistency: the model’s distance comparisons are mutually contradictory.

Angular sector system $\mathcal{P}_{\text{ang}}$. Each TRR response specifies a clock-hour direction for the target relative to a reference frame defined by two other objects. This maps to an angular arc interval: an hour-level response (30° sector) constrains the relative angle α (Eq. 1) to lie within $\pm 15^\circ$ of the expected direction. When multiple TRR responses constrain the same object triple, the feasible region is the intersection of their arcs; an empty intersection indicates conflicting directional judgments.

Constraint extraction modes. The benchmark supports two extraction modes. In the *ground-truth* mode, only correctly answered probes contribute constraints, testing whether accurate sub-answers suffice for global reconstruction. In the *belief* mode, all model predictions (correct or not) are used, reconstructing the model’s *externalized spatial belief* regardless of correctness—analogous to how behavioral experiments infer latent representations from response patterns rather than accuracy alone.

4.2 Optimization

Given the constraint sets $\mathcal{P}_{\text{dist}}$ and $\mathcal{P}_{\text{ang}}$, the benchmark solves for 2D object positions $\{(x_i, y_i)\}_{i=1}^n$ that best satisfy all constraints. The optimization uses L-BFGS-B with a gauge-fixing scheme that removes the inherent ambiguities of ordinal data.

Gauge fixing. Ordinal constraints determine positions only up to a similarity transformation (translation, rotation, scale) and reflection. To resolve these five degrees of freedom, three *anchor* objects are selected by constraint participation frequency and fixed:

$$\mathbf{x}_a = (0, 0), \quad \mathbf{x}_b = (1, 0), \quad y_c \geq 0. \quad (2)$$

The first two anchors fix translation, rotation, and scale (4 DOF); the inequality on y_c breaks the reflection ambiguity (1 DOF). The remaining $2(n-2)$ coordinates are free variables optimized via L-BFGS-B, with $y_c \geq 0$ enforced as a box constraint.

Loss function. The total objective combines three terms: $\mathcal{L} = \mathcal{L}_{\text{QRR}} + \mathcal{L}_{\text{TRR}} + \mathcal{L}_{\text{sep}}$.

For QRR constraints, we use a log-domain ranking loss. Let $\delta = \log d(A, B) - \log d(C, D)$. For a “<” constraint, the loss is a softplus penalty when the margin is violated:

$$\mathcal{L}_{\text{QRR}}^{(<)} = \frac{1}{\beta} \log(1 + \exp(\beta(\delta + m))), \quad (3)$$

where $m = 0.1$ is the margin and $\beta = 10$ controls sharpness. The “>” case is symmetric ($-\delta + m$), and “ $\approx$ ” uses a Huber loss on δ with tolerance $\delta_{\text{eq}} = 0.1$. The log domain ensures scale invariance: the loss depends only on distance *ratios*, matching the ordinal nature of QRR comparisons.

For TRR constraints, we use a sector-tolerance loss. Let $\mathbf{u}$ be the unit vector from **ref1** to **ref2** (the 12 o’clock direction), $\mathbf{v}$ the unit vector from **ref1** to the target, and $\mathbf{u}_\alpha = R(\alpha)\mathbf{u}$ the expected direction for clock hour h (where $\alpha = 30^\circ \cdot (h \bmod 12)$). The loss penalizes configurations where the cosine similarity falls below the sector boundary:

$$\mathcal{L}_{\text{TRR}} = \frac{1}{\beta} \log(1 + \exp(\beta \cdot \frac{\cos \theta_{\text{tol}} - \mathbf{u}_\alpha \cdot \mathbf{v}}{\tau_s})), \quad (4)$$

where $\theta_{\text{tol}} = 15^\circ$ for hour-level and 45° for quadrant-level constraints, and $\tau_s = 0.1$ is a softening temperature (distinct from the QRR comparator tolerance τ in §3.2).

A separation regularizer $\mathcal{L}_{\text{sep}} = \lambda \sum_{i < j} \max(0, \epsilon - \|\mathbf{x}_i - \mathbf{x}_j\|)$ with $\epsilon = 0.05$ and $\lambda = 1.0$ prevents object collapse.

Multi-start optimization. Because the loss landscape may contain local minima (especially for complex scenes or inconsistent constraints), the solver runs $K = 10$ restarts from random initializations and collects all converged solutions, sorted by loss. The diversity of solutions is itself informative: a unique minimum suggests the constraints tightly determine the layout, while multiple distinct minima indicate geometric ambiguity.

4.3 Scene-Level Metrics

The reconstruction produces a suite of scene-level metrics that go beyond per-question accuracy.

Constraint Satisfaction Rate (CSR). The fraction of input constraints satisfied by the best reconstruction. CSR_{QRR} checks whether the reconstructed pairwise distances preserve each ordinal comparison (using the same $\tau = 0.10$ tolerance); CSR_{TRR} checks whether reconstructed angles fall within the specified sector. A CSR near 1.0 means the model’s answers are mutually consistent; a low CSR means no single layout can reconcile them.

Normalized RMS Error (NRMS). When ground-truth positions are available, the best reconstruction is Procrustes-aligned (translation, rotation, uniform scale) to the ground truth, and the root-mean-square positional error is normalized by the spatial extent of the ground-truth layout:

$$\text{NRMS} = \frac{\text{RMS}(\hat{\mathbf{X}}_{\text{aligned}}, \mathbf{X}_{\text{GT}})}{\max_k \text{range}_k(\mathbf{X}_{\text{GT}})}, \quad (5)$$

where range_k denotes the extent along coordinate axis k . $\text{NRMS} \approx 0$ indicates a faithful reconstruction; $\text{NRMS} > 0.3$ typically indicates severe geometric distortion.

Kendall τ . The Kendall rank correlation between reconstructed and ground-truth pairwise distances measures how well the ordinal distance structure is preserved, independent of metric accuracy. A τ near 1.0 means the model preserves the “which pair is closer” ordering; a τ near 0 indicates no ordinal structure.

Geometric multiplicity (K_{geom}) and spread. From the multi-start solutions, we cluster converged results by RMS distance (threshold 0.10) after filtering solutions with loss exceeding $3 \times$ the best. K_{geom} is the number of distinct clusters (geometric modes). A *spread* metric quantifies the average RMS distance of good solutions from the best, measuring how tightly the constraints pin down the layout. $K_{\text{geom}} = 1$ with low spread indicates a well-determined reconstruction; $K_{\text{geom}} > 1$ indicates the constraints admit multiple qualitatively different layouts.

4.4 Validation

To verify the reconstruction pipeline, it is run on ground-truth constraints extracted directly from the scene annotations (i.e., all QRR and TRR answers are correct). Under these ideal inputs, the solver consistently recovers layouts with $\text{CSR}_{\text{QRR}} \approx 1.0$, $\text{NRMS} \approx 0.01$, and $K_{\text{geom}} = 1$ across scenes of all complexity levels ($n = 4$ to $n = 10$). This confirms that the loss functions and gauge-fixing scheme faithfully recover the ground-truth geometry when the input constraints are consistent, establishing a sound baseline for interpreting VLM reconstruction results: any degradation in reconstruction quality can be attributed to errors or inconsistencies in the model’s responses, not to limitations of the reconstruction method itself.

5 Separating Vision from Language

Question-level accuracy alone cannot tell us whether a model’s spatial answers are genuinely grounded in the image or are largely recoverable from object names, task format, and generic priors. Ordinary-Bench therefore includes a second layer of analysis that explicitly separates visual contribution from language-driven guessing. The benchmark is designed so that the same scene, the same probe set, and the same scoring pipeline can be evaluated under controlled perturbations of the visual input.

5.1 Three-Condition Protocol

Section 3.3 defined the three evaluation conditions: correct image, mismatched image, and no image. Their role is not merely ablative, but diagnostic. Condition A asks whether the model can use the appropriate visual evidence. Condition B tests whether the model is sensitive to image-content mismatch rather than simply exploiting the presence of *some* image. Condition C estimates how much of the task can be solved from textual priors alone.

This setup yields a scene-level measure of visual grounding. Let NRMS_A , NRMS_B , and NRMS_C denote reconstruction error under the three conditions. We define

$$\text{VIG} = \text{NRMS}_C - \text{NRMS}_A, \tag{6}$$

so that a positive score indicates that the correct image leads to a better recovered scene than text-only prompting. We also track the gap between mismatched and correct images, $\text{NRMS}_B - \text{NRMS}_A$,

to test whether the model is using scene-specific visual evidence rather than merely benefiting from multimodal formatting. In the final evaluation, these quantities should be paired with scene-wise significance tests such as permutation tests or paired nonparametric tests.

5.2 Consistency as Structural Evidence

Visual grounding should improve not only accuracy but also the internal structure of a model’s answers. We therefore complement VIG with consistency diagnostics that operate directly on the predicted constraint set.

Transitivity for QRR. Distance comparisons admit natural transitive checks. If a model predicts $d(A, B) < d(C, D)$ and $d(C, D) < d(E, F)$, then a coherent ordinal belief should also support $d(A, B) < d(E, F)$. We summarize violations with a *transitivity violation rate*,

$$T_{\text{viol}} = \frac{\#\{\text{violated testable transitive implications}\}}{\#\{\text{testable transitive implications}\}}. \quad (7)$$

Low transitivity violation indicates that the model’s distance judgments are compatible with a single partial order rather than a collection of unrelated local guesses.

Reciprocity for TRR. Directional judgments admit an analogous reciprocity check. If a target is judged to lie at 3 o’clock in the reference frame defined by (**ref1**, **ref2**), then swapping the viewing direction should induce the corresponding opposite direction. We summarize such failures with a *reciprocity violation rate* R_{viol} . These structural metrics are especially informative because purely language-driven heuristics may occasionally answer individual probes correctly while still failing to satisfy the geometric relations that tie probes together.

Selective answering. The benchmark is also compatible with selective-answering variants in which the model may abstain, return a coarse answer, or express ambiguity when the visual evidence is weak. In that setting, abstention is not treated as failure by default; rather, it can improve the quality of the extracted constraint set by preventing unreliable observations from contaminating reconstruction. This follows the broader perspective that reliable multimodal evaluation should distinguish “cannot tell” from “guessed incorrectly” [Whitehead et al. \[2022\]](#), [Eisenschlos et al. \[2024\]](#), [Khan and Fu \[2024\]](#).

5.3 Random Baselines and Belief Faithfulness

To calibrate both accuracy and reconstruction quality, we compare model behavior against random baselines matched to task arity: 1/3 for QRR, 1/12 for TRR hour-level accuracy, and 1/4 for TRR quadrant accuracy. However, random-answer baselines are not sufficient on their own, because the reconstruction solver may impose weak geometric regularities even on noisy constraint sets. We therefore recommend reporting a *belief faithfulness gap*: the difference between the model’s reconstruction quality and that of a random baseline under the same solver, metrics, and scene set.

This comparison matters for interpretation. A model with only modest question-level gains over random may still produce a substantially more coherent global belief, while a model with apparently non-trivial accuracy may collapse to near-random behavior once its answers are examined structurally. Ordinary-Bench is intended to make this distinction explicit.

6 Experiments

We organize the experimental analysis of Ordinary-Bench along four axes: question-level accuracy, scene belief reconstruction, visual contribution separation, and structural consistency. This section provides the evaluation setup and the reporting structure used by the benchmark. Final numerical results, tables, and figures can be filled in once the full model roster and scene coverage are frozen.

6.1 Experimental Setup

Models. The evaluation pipeline is model-agnostic and accepts any OpenAI-compatible endpoint through environment variables, making it straightforward to compare proprietary APIs and open-weight models behind a unified prompting and scoring stack. The submission version should cover multiple model families, ideally including both closed-source systems (e.g., GPT-4o, Gemini 2.0 Flash, Claude-class models) and open or open-weight systems (e.g., Qwen-VL, LLaVA-OneVision, InternVL-class models), so that the conclusions are not tied to a single provider family.

Prompting and inference. All experiments use structured batch prompting with machine-parseable JSON outputs. By default, questions are grouped in batches of 20, decoded at temperature 0, and retried through a short ReAct-style correction loop when the model returns malformed or incomplete responses. This protocol is implemented directly in the evaluation code and is shared across single-view and multi-view settings.

Scene coverage and splits. The benchmark contains seven complexity splits corresponding to scenes with $n = 4, \dots, 10$ objects. We report metrics both aggregated across all scenes and broken down by object count to expose performance degradation as combinatorial relation density increases. The repository also supports train/test scene partitions for development without test leakage.

Metrics. At the question level, we report QRR exact accuracy and TRR accuracy at hour, quadrant, and adjacent-hour granularities. At the scene level, we report CSR, NRMS, Kendall τ , K_{geom} , and spread. For the visual-separation protocol, we report NRMS_A , NRMS_B , NRMS_C , VIG, and consistency measures such as T_{viol} and R_{viol} .

6.2 Question-Level Accuracy

The first analysis reports conventional accuracy metrics as a function of scene complexity. The goal is to establish how well each model answers local ordinal probes before any scene-level interpretation is applied. We expect this view to be necessary but insufficient: models may exhibit non-trivial QRR or coarse TRR performance while still failing to maintain scene-level coherence.

The final paper should include a table of aggregate accuracies and a complexity curve showing how QRR and TRR performance degrade from $n = 4$ to $n = 10$. A random baseline and, if available, a small human baseline should be shown on the same plots for calibration.

6.3 Scene Belief Reconstruction

The second analysis asks whether local answers can be assembled into a single coherent layout. For each scene and model, we reconstruct the best-fitting scene belief from predicted constraints and

evaluate it with CSR, NRMS, Kendall τ , K_{geom} , and spread. These metrics expose three failure modes that accuracy alone cannot separate: mutually inconsistent answers, geometrically distorted but roughly ordered beliefs, and underconstrained multimodal beliefs.

The final version should include both aggregate statistics and qualitative visualizations of reconstructed layouts. Particularly useful figures include matched examples comparing ground-truth layouts with reconstructions from strong and weak models, as well as distribution plots over NRMS and CSR across scenes.

6.4 Visual Contribution Separation

The third analysis applies the three-condition protocol to the same question sets and compares reconstruction quality across correct-image, mismatched-image, and no-image settings. The central quantity is VIG, which measures how much scene reconstruction improves when the model receives the correct image instead of text alone. A strong positive VIG indicates genuine visual contribution; near-zero improvement suggests that the model’s spatial belief is dominated by textual priors.

In the final paper, this analysis should be reported both numerically and visually. A table of NRMS_A , NRMS_B , NRMS_C , and VIG per model should be paired with qualitative examples showing how the same scene belief changes across conditions.

6.5 Consistency Analysis

The fourth analysis measures whether a model’s answers obey the structural regularities implied by the tasks themselves. For QRR, this means transitivity over distance comparisons; for TRR, reciprocity over directional judgments. These metrics help distinguish a model that truly preserves spatial structure from one that simply happens to answer some probes correctly in isolation.

The final report should compare consistency rates across models and across the three visual conditions. If condition C already exhibits high answer rates but poor structural consistency, that would provide direct evidence that language priors can mimic competence at the local level without supporting a coherent scene belief.

7 Discussion and Limitations

What VLMs seem to see. Ordinary-Bench is designed to test a narrower and more diagnostic claim than general multimodal competence. A model may identify objects correctly, produce plausible captions, or answer some spatial prompts accurately while still failing to maintain a stable geometric belief over the full scene. The key distinction is therefore not between “correct” and “incorrect” in the abstract, but between local success and globally coherent, visually grounded structure.

Why scene-level analysis matters. This distinction is important because many downstream uses of VLMs implicitly assume more than isolated correctness. Planning, embodied interaction, and multi-step reasoning all require judgments that are mutually compatible. If a model says that one pair of objects is closer than another, and separately answers directional probes about the same scene, those answers should support a single world rather than a patchwork of unrelated guesses. Scene belief reconstruction makes this compatibility test explicit.

Why controlled synthetic data is a feature here. The benchmark deliberately uses controlled rendered scenes instead of unconstrained real-world imagery. This lowers ecological realism, but it makes exact geometry, dense probing, and visual perturbation possible. For the scientific question posed here, this tradeoff is appropriate: it is difficult to separate visual evidence from language priors, or to measure scene-level coherence, when the underlying scene structure is itself ambiguous. Controlled synthetic data therefore acts as an instrument for diagnosis rather than as a claim of direct real-world coverage.

Limitations. The current benchmark has several limitations. First, the core reconstruction pipeline operates in 2D or 2.5D terms even though scenes are rendered from 3D environments; this matches the semantics of the current TRR implementation but does not amount to full 3D scene recovery. Second, the main evaluation setting treats models as API-level black boxes, so we analyze externalized behavior rather than hidden representations. Third, the benchmark focuses on simple geometric objects in synthetic scenes; transferring conclusions to natural images or embodied environments requires additional validation. Finally, the difficulty of fine-grained TRR prediction means that some conclusions may be clearer at coarser directional granularity than at exact clock-hour resolution.

Future directions. The natural next steps are to complete the temporal extension already prototyped in the repository, expand the release package for broader community use, and connect behavioral reconstruction with internal representation analysis for open-weight models. Another promising direction is to combine selective answering with richer output forms such as set-valued or graph-structured scene descriptions, which could yield cleaner constraint extraction than forced single-label responses.

8 Release, Reproducibility, and Ethics

Release plan. The repository already contains the core generation and evaluation pipeline: Blender-based scene generation, question enumeration, VLM API evaluation, response parsing, scoring, and reconstruction analysis. The benchmark release package should include rendered images, scene JSON files, split indices, generated question files, prompting templates, evaluation scripts, and aggregate analysis utilities. The current paper draft also assumes a dataset card and machine-readable metadata package (e.g., Croissant-style metadata) will be added before submission or public release.

Reproducibility. Ordinary-Bench is fully programmatic: scene annotations are derived from rendered geometry rather than manual labeling, and question generation is deterministic given the scene metadata and tolerance settings. This substantially reduces annotation ambiguity and makes it possible to regenerate the benchmark from source. For model evaluation, the codebase standardizes endpoint configuration, batching, retry behavior, JSON parsing, and scoring, so that comparisons across models remain protocol-consistent. To improve reproducibility further, the release should freeze prompt templates, random seeds where applicable, and exact metric definitions used in the final tables.

Licensing and access. The release should separate code and data licensing clearly. A practical target is an open-source code license for the pipeline and a permissive data license for rendered scenes and annotations, subject to final confirmation. Closed-source models evaluated through APIs may require paid access and external credentials, so exact reproduction of all leaderboard entries may not be possible without matching provider access.

Compute and cost. The benchmark incurs two main resource costs: rendering and API evaluation. Rendering cost scales with the number of scenes, Cycles sample count, and number of views. Evaluation cost scales with model pricing, concurrency, retry settings, and the total number of probe batches. Reporting these costs is important for a Datasets & Benchmarks submission because they shape the practical accessibility of the benchmark for follow-up work.

Ethical considerations. The current benchmark uses synthetic scenes composed of simple objects and contains no people, faces, or sensitive real-world imagery. This sharply limits privacy and content-moderation concerns relative to web-scale scraped datasets. At the same time, the benchmark should not be interpreted as a comprehensive measure of real-world visual reasoning, and any leaderboard use should avoid overstating what performance on controlled geometric scenes implies about broader competence or safety.

9 Conclusion

Ordinary-Bench is built around a simple claim: evaluating spatial intelligence in vision-language models requires more than counting correct answers. By combining controlled 3D-rendered scenes, dense ordinal probing, scene belief reconstruction, and explicit visual-ablation conditions, the benchmark turns scene-level coherence and visual grounding into measurable quantities. This shifts evaluation from “how many questions did the model answer correctly?” toward “what kind of scene belief does the model actually externalize?”

That shift is the main contribution of the benchmark. Even in visually simple, human-intuitive scenes, a model can appear locally competent while failing to support a single coherent geometric interpretation. Ordinary-Bench is designed to make that gap visible and reproducible.

References

- Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Dorsa Sadigh, Leonidas Guibas, and Fei Xia. Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024a. URL https://openaccess.thecvf.com/content/CVPR2024/papers/Chen_SpatialVLM_Endowing_Vision-Language_Models_with_Spatial_Reasoning_Capabilities_CVPR_2024_paper.pdf.
- Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. Blink: Multimodal large language models can see but not perceive. *arXiv preprint arXiv:2404.12390*, 2024a. URL <https://arxiv.org/abs/2404.12390>.
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. Are we on the right way for evaluating large

- vision-language models? *arXiv preprint arXiv:2403.20330*, 2024b. URL <https://arxiv.org/abs/2403.20330>.
- Kang-il Lee, Minbeom Kim, Seunghyun Yoon, Minsung Kim, Dongryeol Lee, Hyukhun Koh, and Kyomin Jung. Vlind-bench: Measuring language priors in large vision-language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025. URL <https://aclanthology.org/2025.findings-naacl.231/>.
- Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, Peixian Chen, Yanwei Li, Shaohui Lin, Sirui Zhao, Ke Li, Tong Xu, Xiwu Zheng, Enhong Chen, Caifeng Shan, Ran He, and Xing Sun. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024b. URL <https://arxiv.org/abs/2405.21075>.
- Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Zhengyang Liang, Shitao Xiao, Minghao Qin, Xi Yang, Yongping Xiong, Bo Zhang, Tiejun Huang, and Zheng Liu. Mlvu: Benchmarking multi-task long video understanding. *arXiv preprint arXiv:2406.04264*, 2024. URL <https://arxiv.org/abs/2406.04264>.
- Yuanxin Liu, Shicheng Li, Yi Liu, Yuxiang Wang, Shuhuai Ren, Lei Li, Sishuo Chen, Xu Sun, and Lu Hou. Tempcompass: Do video llms really understand videos? *arXiv preprint arXiv:2403.00476*, 2024. URL <https://arxiv.org/abs/2403.00476>.
- Shijie Zhou, Alexander Vilesov, Xuehai He, Ziyu Wan, Shuwang Zhang, Aditya Nagachandra, Di Chang, Dongdong Chen, Xin Eric Wang, and Achuta Kadambi. Vlm4d: Towards spatiotemporal awareness in vision language models. *arXiv preprint arXiv:2508.02095*, 2025. URL <https://arxiv.org/abs/2508.02095>.
- Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick. CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2901–2910, 2017. URL <https://arxiv.org/abs/1612.06890>.
- Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B. Tenenbaum. CLEVRER: Collision events for video representation and reasoning. In *International Conference on Learning Representations*, 2020. URL <https://arxiv.org/abs/1910.01442>.
- Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008. URL <https://doi.org/10.3389/neuro.06.004.2008>.
- Daniel L. K. Yamins and James J. DiCarlo. Using goal-driven deep learning models to understand sensory cortex. *Nature Neuroscience*, 19(3):356–365, 2016. URL <https://doi.org/10.1038/nn.4244>.
- Spencer Whitehead, Vedaad Shakib, Suzanne Petryk, and Marcus Rohrbach. Reliable visual question answering: Abstain rather than answer incorrectly. *arXiv preprint arXiv:2210.07000*, 2022. URL <https://arxiv.org/abs/2210.07000>.
- Julian Martin Eisenschlos, Hernan Maina, Guido Ivetta, and Luciana Benotti. Selectively answering visual questions. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024. URL <https://aclanthology.org/2024.findings-acl.20/>.

- Zaid Khan and Yun Fu. Consistency and uncertainty: Identifying unreliable responses from black-box vision-language models for selective visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Khan_Consistency_and_Uncertainty_Identifying_Unreliable_Responses_From_Black-Box_Vision-Language_Models_CVPR_2024_paper.html.
- Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. In *Transactions of the Association for Computational Linguistics*, volume 11, pages 635–651, 2023. URL <https://arxiv.org/abs/2205.00363>.
- Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? investigating their struggle with spatial reasoning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. URL <https://arxiv.org/abs/2310.19785>.
- Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. URL <https://arxiv.org/abs/2305.10355>.
- Sameer Agarwal, Josh Wills, Lawrence Cayton, and Gert Lanckriet. Generalized non-metric multidimensional scaling. In *Proceedings of the Eleventh International Conference on Artificial Intelligence and Statistics*, 2007. URL <https://proceedings.mlr.press/v2/agarwal07a.html>.
- Ke Ma, Qianqian Xu, and Xiaochun Cao. Robust ordinal embedding from contaminated relative comparisons. *arXiv preprint arXiv:1812.01945*, 2018. URL <https://arxiv.org/abs/1812.01945>.
- Ivan Dokmanić, Reza Parhizkar, Juri Ranieri, and Martin Vetterli. Euclidean distance matrices: Essential theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 32(6):12–30, 2015. URL <https://arxiv.org/abs/1502.07541>.
- Shiyu Zhao and Daniel Zelazo. Bearing rigidity theory and its applications for control and estimation of network systems: Life beyond distance rigidity. *IEEE Control Systems Magazine*, 39(2):66–83, 2019. URL <https://eprints.whiterose.ac.uk/127410/>.
- Yiguang Chen, Ming Cao, and Chi Li. Angle rigidity and its usage to stabilize multi-agent formations in 2-d. *IEEE Transactions on Automatic Control*, 66(11):5417–5423, 2021. URL https://pure.rug.nl/ws/portalfiles/portal/177665656/Angle_Rigidity_and_Its_Usage_to_Stabilize_Multiagent_Formations_in_2_D_1_.pdf.